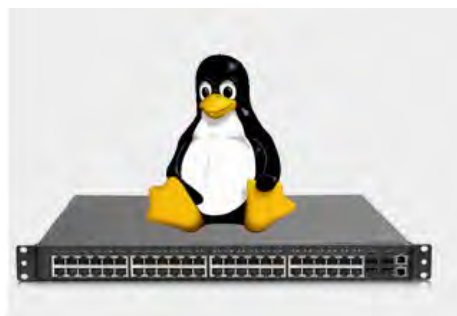




Configuración de la red en Linux

Alfredo Abad

PARP405-LinuxConfigRed.pptx

10-sep-2024

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Objetivos y escenario

- **Objetivos:**
 - Aprender a configurar la red en sistemas Linux por diversos métodos y sobre dos plataformas (Desktop y Server)
 - Estudiar las posibilidades de configuración de:
 - **Network Manager** sobre Linux Desktop
 - El fichero **/etc/network/interfaces**
 - Configuración a través de **NetPlan**
- **Escenario:**
 - Sobre un Linux Desktop con una única tarjeta de red (comprueba que navegas)
 - Configurar la red mediante Network Manager para que tome dirección IP dinámica (IP, máscara, Gateway, DNS)
 - Repetir la operación con IP estática
 - Sobre un Linux Server con dos tarjetas de red (comprueba que navegas)
 - Configurar las dos tarjetas con NetPlan (ver detalle en diapo siguiente)
 - Repetir el proceso con /etc/network/interfaces

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Pruebas de configuración sobre Linux Server (debe probarse lo siguiente)

- Configurar una tarjeta de red con IP dinámica y la otra con IP estática
- Prueba a asignar la puerta por defecto en una de las tarjetas y luego en la otra
- Crea un bridge entre la dos tarjetas y configura en él:
 - Una IP dinámica (probar)
 - Una IP estática (probar)

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Para entregar

- Prueba de configuración (imagen de ifconfig o ip, fichero de configuración, prueba de navegación, etc.) de cada uno de los escenarios
- Escenarios:
 - A. Linux Desktop con DHCP
 - B. Linux Desktop con IP estática
 - C. Linux Server con NetPlan
 - D. Linux Server con /etc/network/interfaces
 - E. Linux Server con Netplan y bridge
 - F. Linux Server con /etc/network/interfaces y bridge

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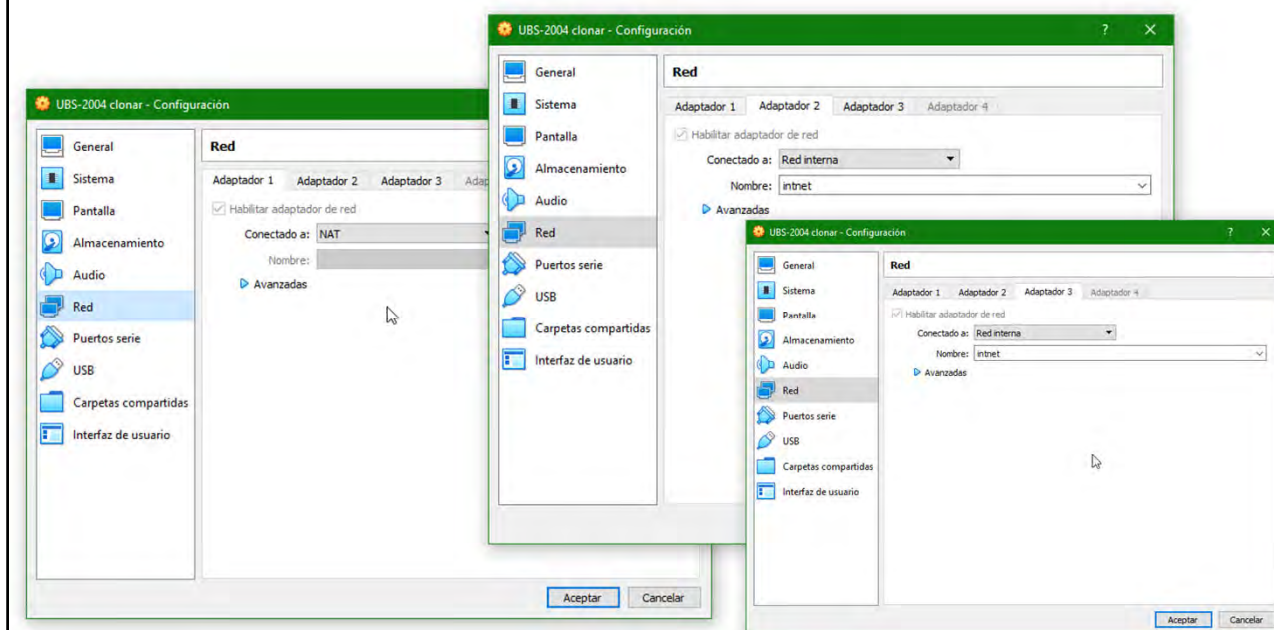


Ejemplos de operación con Netplan

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Tres NIC: una en NAT (dhcp) y dos más (red interna)



Vista inicial de la red: lo, enp0s3, enp0s8 y enp0s9

```

root@ubs:/home/labadmin# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:44:15:49 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86226sec preferred_lft 86226sec
    inet6 fe80::a00:27ff:fe44:1549/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:ca:98:04 brd ff:ff:ff:ff:ff:ff
4: enp0s9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:d0:f6:cf brd ff:ff:ff:ff:ff:ff
root@ubs:/home/labadmin#
  
```

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Vista del fichero netplan inicial del que se parte (en este caso solo la red NAT está activa y por dhcp)

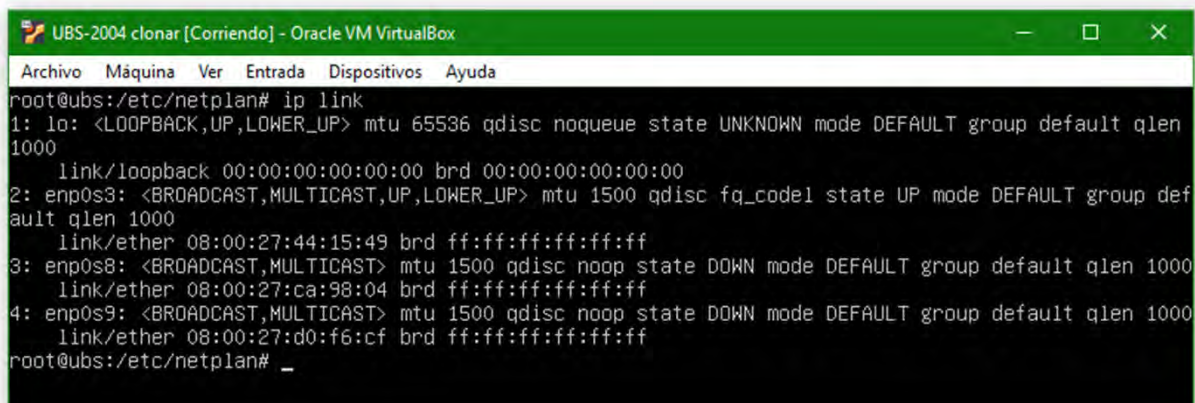
```

root@ubs:/home/labadmin# ls /etc/netplan
00-installer-config.yaml
root@ubs:/home/labadmin# cd /etc/netplan
root@ubs:/etc/netplan# cat 00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      version: 2
root@ubs:/etc/netplan#
  
```

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Vista de las interfaces de red al inicio (sin configurar)

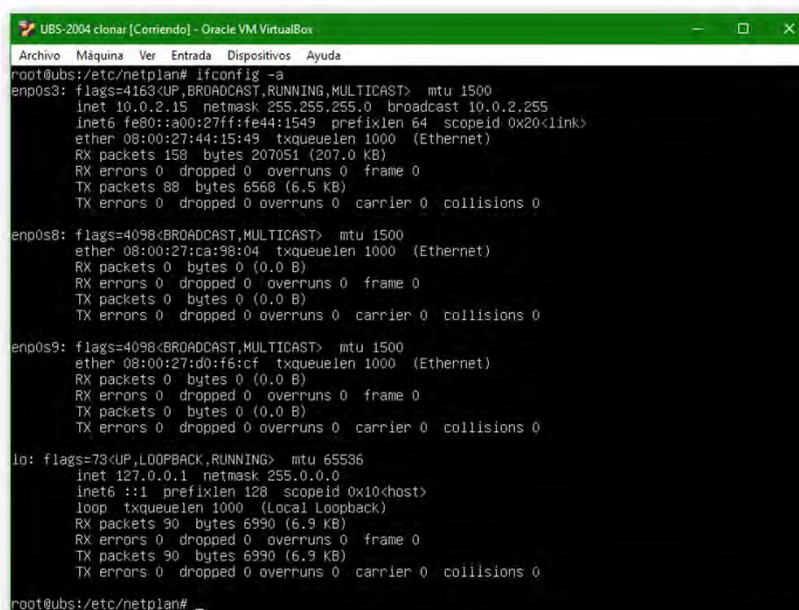


```
UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_code1 state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:44:15:49 brd ff:ff:ff:ff:ff:ff
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:ca:98:04 brd ff:ff:ff:ff:ff:ff
4: enp0s9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:d0:f6:cf brd ff:ff:ff:ff:ff:ff
root@ubs:/etc/netplan# _
```

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Se instaló net-tools para tener ifconfig (vista inicial)



```
UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# ifconfig -a
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe44:1549 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 150 bytes 207051 (207.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 88 bytes 6568 (6.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 08:00:27:ca:98:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s9: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 08:00:27:d0:f6:cf txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 30 bytes 6990 (6.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 30 bytes 6990 (6.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
root@ubs:/etc/netplan# _
```

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Reescribimos yaml de netplan para configurar un dhcp y dos interfaces con ips estáticas: generamos y aplicamos

```

UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# cat 00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: false
      addresses:
        - 192.168.1.1/24
    enp0s9:
      dhcp4: false
      addresses:
        - 192.168.2.1/24
  version: 2
root@ubs:/etc/netplan#
root@ubs:/etc/netplan# netplan generate
root@ubs:/etc/netplan#
root@ubs:/etc/netplan# netplan apply
root@ubs:/etc/netplan# _

```

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Vista de la red después de aplicar la configuración

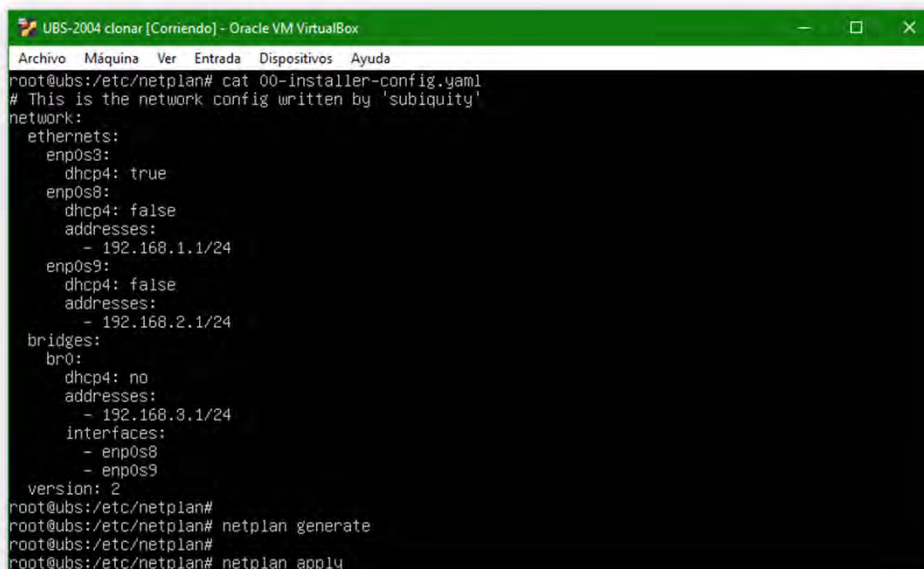
```

UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:44:15:49 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86377sec preferred_lft 86377sec
    inet6 fe80::a00:27ff:fe44:1549/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ca:98:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feca:9804/64 scope link
        valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d0:f6:cf brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.1/24 brd 192.168.2.255 scope global enp0s9
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed0:f6cf/64 scope link
        valid_lft forever preferred_lft forever
root@ubs:/etc/netplan# _

```

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Ahora configuramos un bridge con las dos interfaces estáticas (generamos y aplicamos)



```

UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# cat 00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: false
      addresses:
        - 192.168.1.1/24
    enp0s9:
      dhcp4: false
      addresses:
        - 192.168.2.1/24
  bridges:
    br0:
      dhcp4: no
      addresses:
        - 192.168.3.1/24
      interfaces:
        - enp0s8
        - enp0s9
  version: 2
root@ubs:/etc/netplan# netplan generate
root@ubs:/etc/netplan# netplan apply

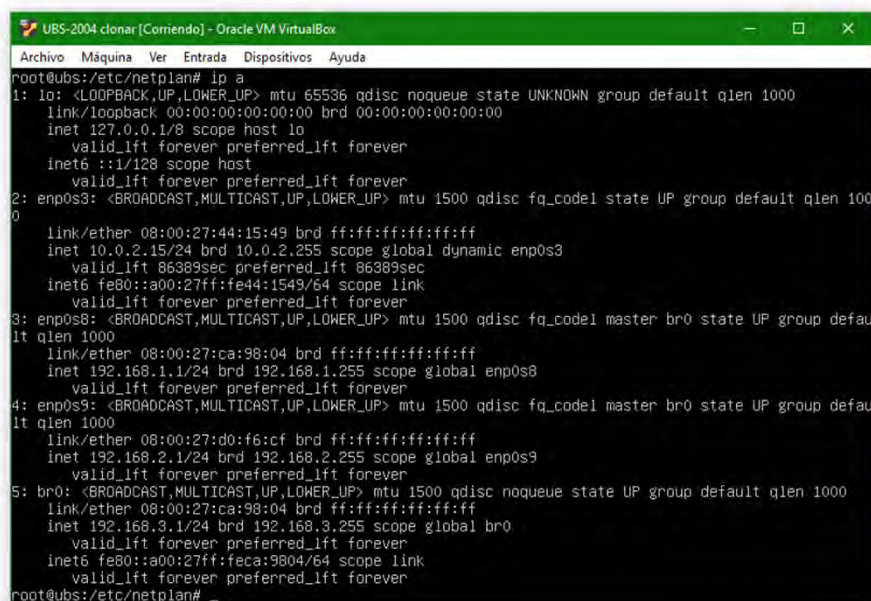
```

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Vista del estado de la red con el nuevo bridge



```

UBS-2004 clonar [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/netplan# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:44:15:49 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86389sec preferred_lft 86389sec
    inet6 fe80::a00:27ff:fe44:1549/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br0 state UP group default qlen 1000
    link/ether 08:00:27:ca:98:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global enp0s8
        valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br0 state UP group default qlen 1000
    link/ether 08:00:27:d0:f6:cf brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.1/24 brd 192.168.2.255 scope global enp0s9
        valid_lft forever preferred_lft forever
5: br0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:ca:98:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.3.1/24 brd 192.168.3.255 scope global br0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feca:9804/64 scope link
        valid_lft forever preferred_lft forever
root@ubs:/etc/netplan#

```

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Ejemplos de operación con ifupdown (ifconfig y /etc/network/interfaces)

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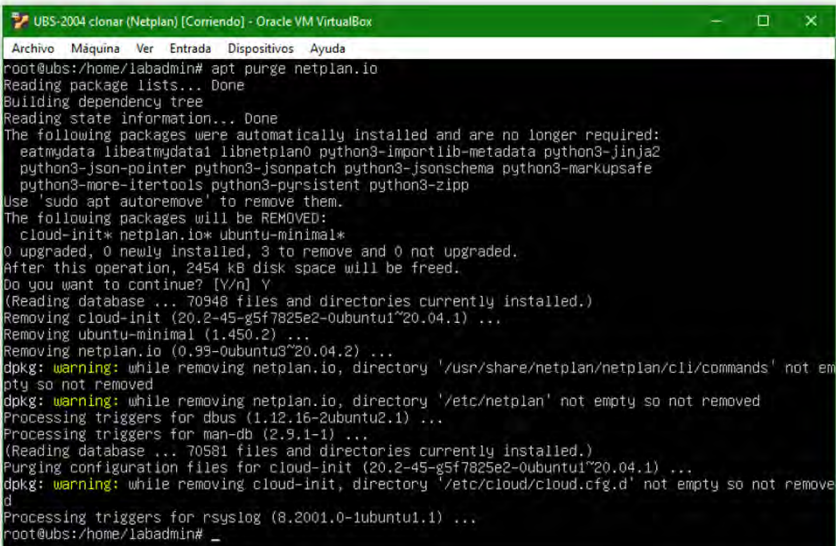
Preparamos sistema instalando net-tools e ifupdown

```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/home/labadmin# apt install net-tools ifupdown
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  pop rdnsd
The following NEW packages will be installed:
  ifupdown net-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 257 kB of archives.
After this operation, 1099 kB of additional disk space will be used.
Get:1 http://es.archive.ubuntu.com/ubuntu focal/universe amd64 ifupdown amd64 0.8.35ubuntu1 [60.5 kB]
Get:2 http://es.archive.ubuntu.com/ubuntu focal/main amd64 net-tools amd64 1.60+git20180626.aebd88e-1ubuntu1 [196 kB]
Fetched 257 kB in 0s (541 kB/s)
Selecting previously unselected package ifupdown.
(Reading database ... 70859 files and directories currently installed.)
Preparing to unpack .../ifupdown_0.8.35ubuntu1_amd64.deb ...
Unpacking ifupdown (0.8.35ubuntu1) ...
Selecting previously unselected package net-tools.
Preparing to unpack .../net-tools_1.60+git20180626.aebd88e-1ubuntu1_amd64.deb ...
Unpacking net-tools (1.60+git20180626.aebd88e-1ubuntu1) ...
Setting up net-tools (1.60+git20180626.aebd88e-1ubuntu1) ...
Setting up ifupdown (0.8.35ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/networking.service + /lib/systemd/system/networking.service.
Created symlink /etc/systemd/system/network-online.target.wants/networking.service + /lib/systemd/system/networking.service.
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for systemd (245.4-4ubuntu3.2) ...
root@ubs:/home/labadmin#
```

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Ahora purgamos netplan.io para que no interfiera y reiniciamos el equipo para que se destruyan todas las configuraciones de red configuradas anteriormente



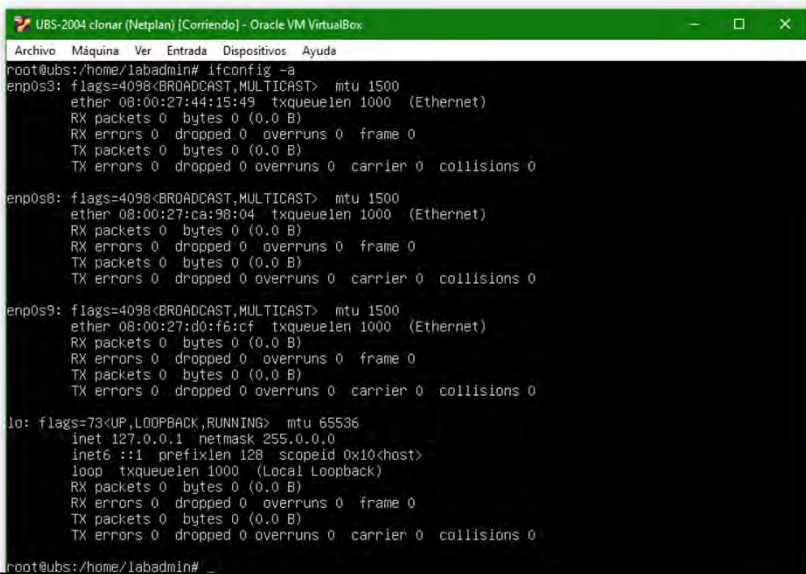
```

UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/home/labadmin# apt purge netplan.io
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  eatmydata libeatmydata libnetplan0 python3-importlib-metadata python3-jinja2
  python3-json-pointer python3-jsonpatch python3-jsonschema python3-markupsafe
  python3-more-itertools python3-pyrsistent python3-zipp
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  cloud-init* netplan.io* ubuntu-minimal*
0 upgraded, 0 newly installed, 3 to remove and 0 not upgraded.
After this operation, 2454 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 70948 files and directories currently installed.)
Removing cloud-init (20.2-45-g5f7825e2-0ubuntu1~20.04.1) ...
Removing ubuntu-minimal (1.450.2) ...
Removing netplan.io (0.99-0ubuntu3~20.04.2) ...
dpkg: warning: while removing netplan.io, directory '/usr/share/netplan/netplan/cli/commands' not empty so not removed
dpkg: warning: while removing netplan.io, directory '/etc/netplan' not empty so not removed
Processing triggers for dbus (1.12.16-2ubuntu2.1) ...
Processing triggers for man-db (2.9.1-1) ...
(Reading database ... 70581 files and directories currently installed.)
Purging configuration files for cloud-init (20.2-45-g5f7825e2-0ubuntu1~20.04.1) ...
dpkg: warning: while removing cloud-init, directory '/etc/cloud/cloud.cfg.d' not empty so not removed
Processing triggers for rsyslog (8.2001.0-1ubuntu1.1) ...
root@ubs:/home/labadmin# _

```

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**Vista de la red con ifconfig -a
(ninguna interfaz está configurada)**



```

UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/home/labadmin# ifconfig -a
enp0s3: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 08:00:27:ca:98:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s9: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 08:00:27:d0:f6:cf txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

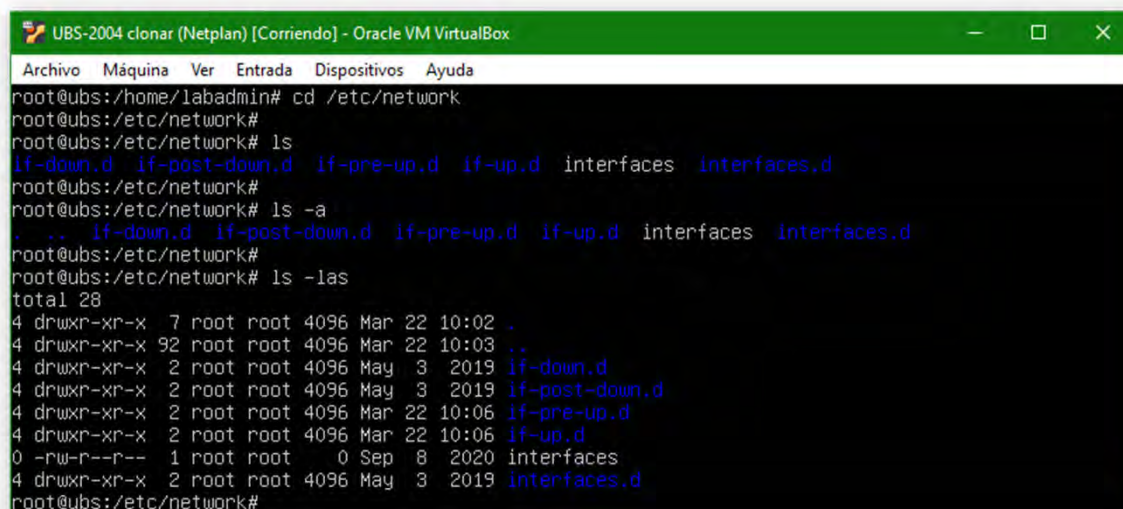
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@ubs:/home/labadmin# _

```

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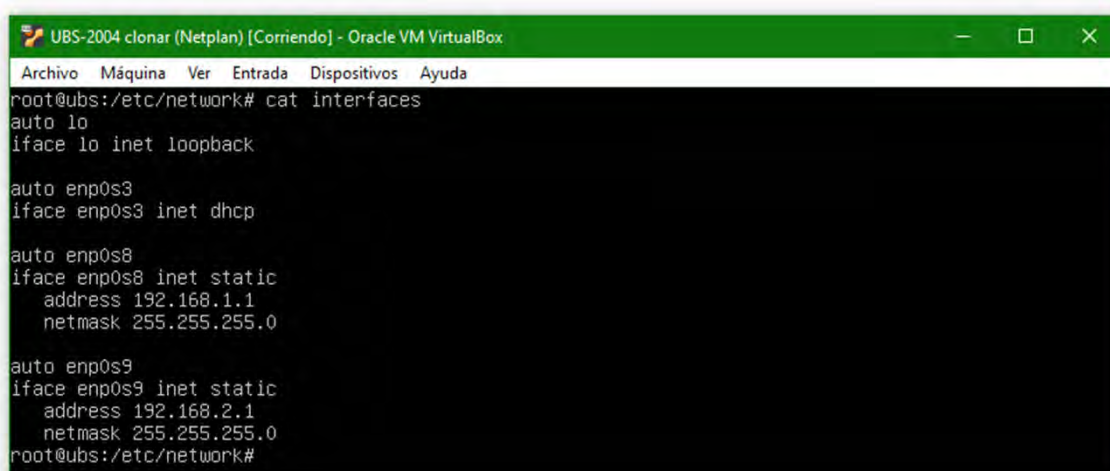
Visualizamos los ficheros de configuración: tenemos fichero /etc/network/interfaces
Si no existiera se crea vacío



```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/home/labadmin# cd /etc/network
root@ubs:/etc/network# ls
if-down.d  if-post-down.d  if-pre-up.d  if-up.d  interfaces  interfaces.d
root@ubs:/etc/network# ls -a
.  ..  if-down.d  if-post-down.d  if-pre-up.d  if-up.d  interfaces  interfaces.d
root@ubs:/etc/network# ls -las
total 28
4 drwxr-xr-x  7 root root 4096 Mar 22 10:02 .
4 drwxr-xr-x 92 root root 4096 Mar 22 10:03 ..
4 drwxr-xr-x  2 root root 4096 May  3 2019 if-down.d
4 drwxr-xr-x  2 root root 4096 May  3 2019 if-post-down.d
4 drwxr-xr-x  2 root root 4096 Mar 22 10:06 if-pre-up.d
4 drwxr-xr-x  2 root root 4096 Mar 22 10:06 if-up.d
0 -rw-r--r--  1 root root    0 Sep  8 2020 interfaces
4 drwxr-xr-x  2 root root 4096 May  3 2019 interfaces.d
root@ubs:/etc/network#
```

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Configuramos la red con una interfaz por dhcp dos más por ip estática (como hicimos con netplan anteriormente)



```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/network# cat interfaces
auto lo
iface lo inet loopback

auto enp0s3
iface enp0s3 inet dhcp

auto enp0s8
iface enp0s8 inet static
    address 192.168.1.1
    netmask 255.255.255.0

auto enp0s9
iface enp0s9 inet static
    address 192.168.2.1
    netmask 255.255.255.0
root@ubs:/etc/network#
```

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Paramos la red con ifdown -a y la volvemos a iniciar con ifup -a

```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/network# ifdown -a
root@ubs:/etc/network# ifup -a
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/enp0s3/08:00:27:44:15:49
Sending on   LPF/enp0s3/08:00:27:44:15:49
Sending on   Socket/fallback
Created uid  "\000\001\000\001\353+\353\010\000'D\025I".
DHCPOFFER of 10.0.2.15 from 10.0.2.2
DHCPREQUEST for 10.0.2.15 on enp0s3 to 255.255.255.255 port 67 (xid=0x2e4d1273)
DHCPACK of 10.0.2.15 from 10.0.2.2 (xid=0x73124d2e)
cmp: EOF on /tmp/tmp.bsk9W0tGX1 which is empty
bound to 10.0.2.15 -- renewal in 34378 seconds.
root@ubs:/etc/network#
```

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Vista de la red configurada con ifconfig

```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe44:1549 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 6 bytes 1620 (1.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 1660 (1.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe44:1549 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 2240 (2.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s9: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.2.1 netmask 255.255.255.0 broadcast 192.168.2.255
    inet6 fe80::a00:27ff:fe44:1549 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 2240 (2.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 644 bytes 45844 (45.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 644 bytes 45844 (45.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

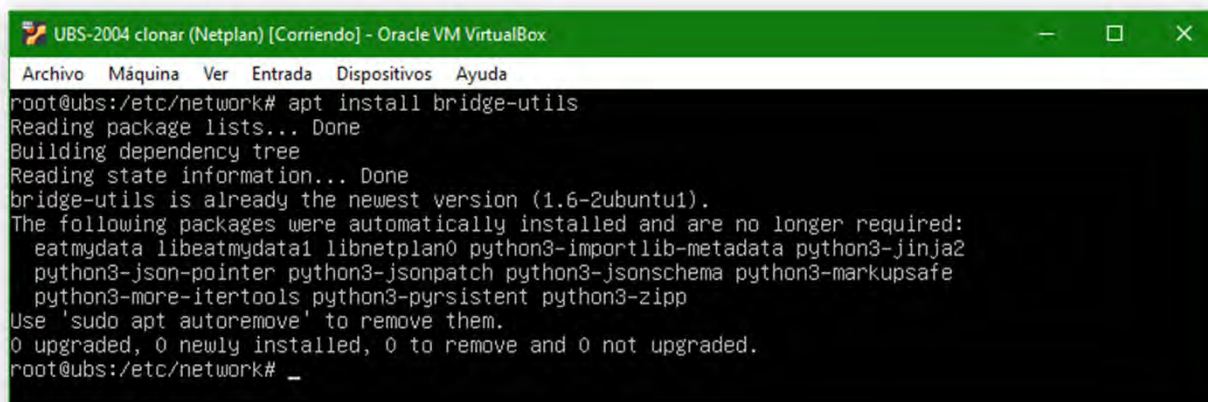
root@ubs:/etc/network#
```

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Para configurar bridges con ifupdown necesitamos las herramientas de bridging:

Las instalamos con `apt install bridge-utils`

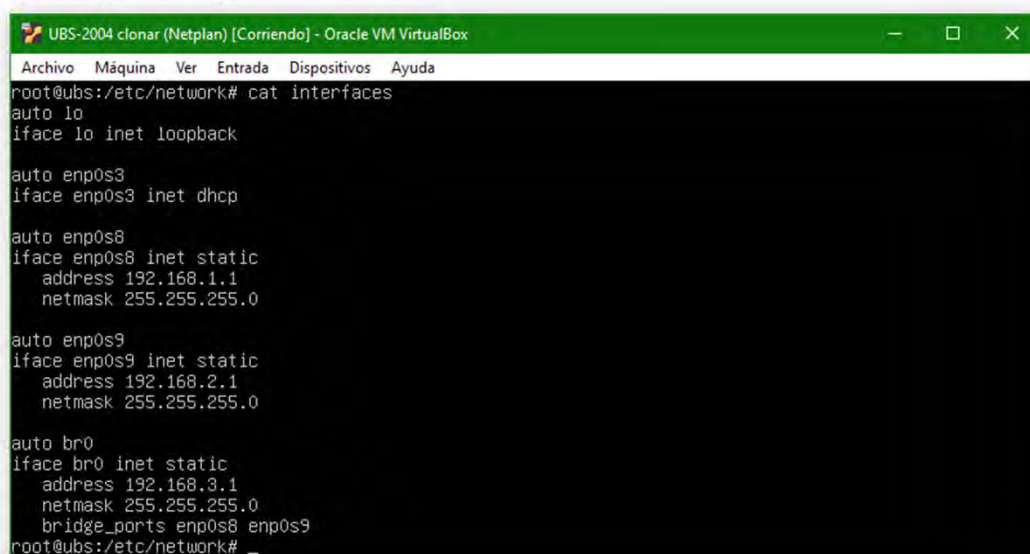


```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/network# apt install bridge-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
bridge-utils is already the newest version (1.6-2ubuntu1).
The following packages were automatically installed and are no longer required:
  eatmydata libeatmydata1 libnetplan0 python3-importlib-metadata python3-jinja2
  python3-json-pointer python3-jsonpatch python3-jsonschema python3-markupsafe
  python3-more-itertools python3-pyrsistent python3-zipp
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ubs:/etc/network# _
```

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Reescribimos `/etc/network/interfaces` para que incluya un bridge (br0) sobre enp0s8 y enp0s9 (como hicimos con netplan). Después reiniciamos la red.



```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
root@ubs:/etc/network# cat interfaces
auto lo
iface lo inet loopback

auto enp0s3
iface enp0s3 inet dhcp

auto enp0s8
iface enp0s8 inet static
    address 192.168.1.1
    netmask 255.255.255.0

auto enp0s9
iface enp0s9 inet static
    address 192.168.2.1
    netmask 255.255.255.0

auto br0
iface br0 inet static
    address 192.168.3.1
    netmask 255.255.255.0
    bridge_ports enp0s8 enp0s9
root@ubs:/etc/network# _
```

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Vista de la red con el bridge configurado

```
UBS-2004 clonar (Netplan) [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
br0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.3.1 netmask 255.255.255.0 broadcast 192.168.3.255
    inet6 fe80::a00:27ff:feea:9804 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:ca:98:04 txqueuelen 1000 (Ethernet)
    RX packets 351128 bytes 25217160 (25.2 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 674 bytes 57972 (57.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe44:1549 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:44:15:49 txqueuelen 1000 (Ethernet)
    RX packets 50 bytes 37713 (37.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 71 bytes 7699 (7.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
    ether 08:00:27:ca:98:04 txqueuelen 1000 (Ethernet)
    RX packets 2230577 bytes 191196910 (191.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2227456 bytes 190908934 (190.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s9: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.2.1 netmask 255.255.255.0 broadcast 192.168.2.255
    ether 08:00:27:d0:f6:cf txqueuelen 1000 (Ethernet)
    RX packets 2227027 bytes 190872372 (190.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2231014 bytes 191234160 (191.2 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
--More--
```

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